

The Collapse of Heterogeneity in Silicon Philosophers

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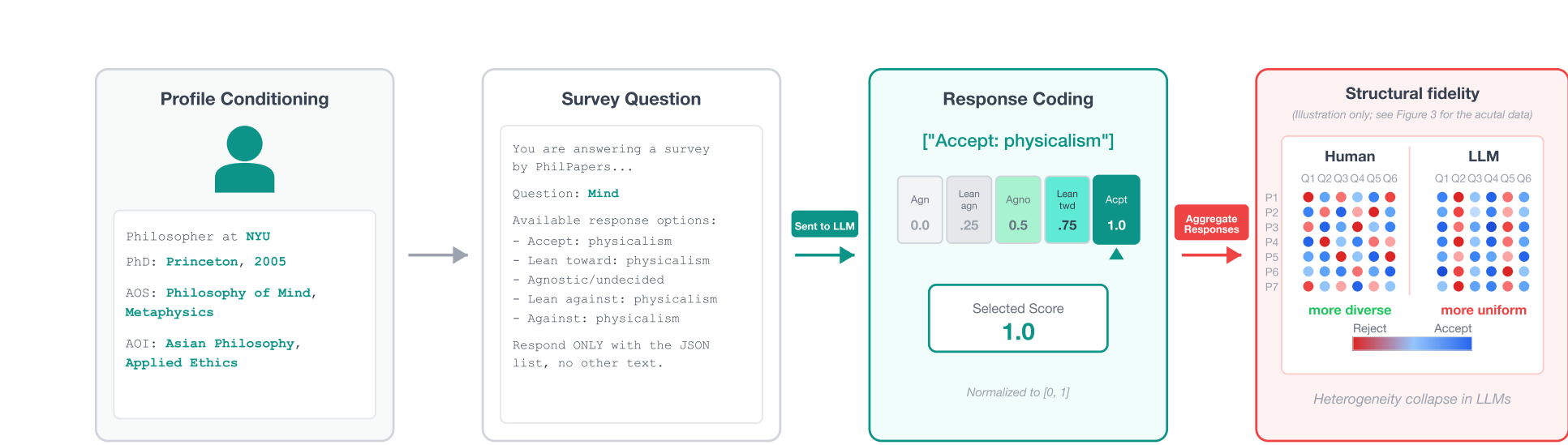
Question

Can an LLM stand in for a population of minds?

Setup

- $N = 277$ PhilPeople philosophers, 100 PhilPapers 2020 questions
- 7 profile-conditioned LLMs, temperature 0
- Accept $\rightarrow$ Reject coded onto $[0, 1]$
- Checked against PhilPapers 2020 ($N = 1,785$)

Silicon sampling



Profile $\rightarrow$ question $\rightarrow$ coded stance $\rightarrow$ fidelity.

How we measure spread

- **Resp%** is how often a source answers a question. Humans skip more; models answer more.
- **Per-Q Var** is the average variance of answers *on the same question*. High variance means philosophers actually disagree.

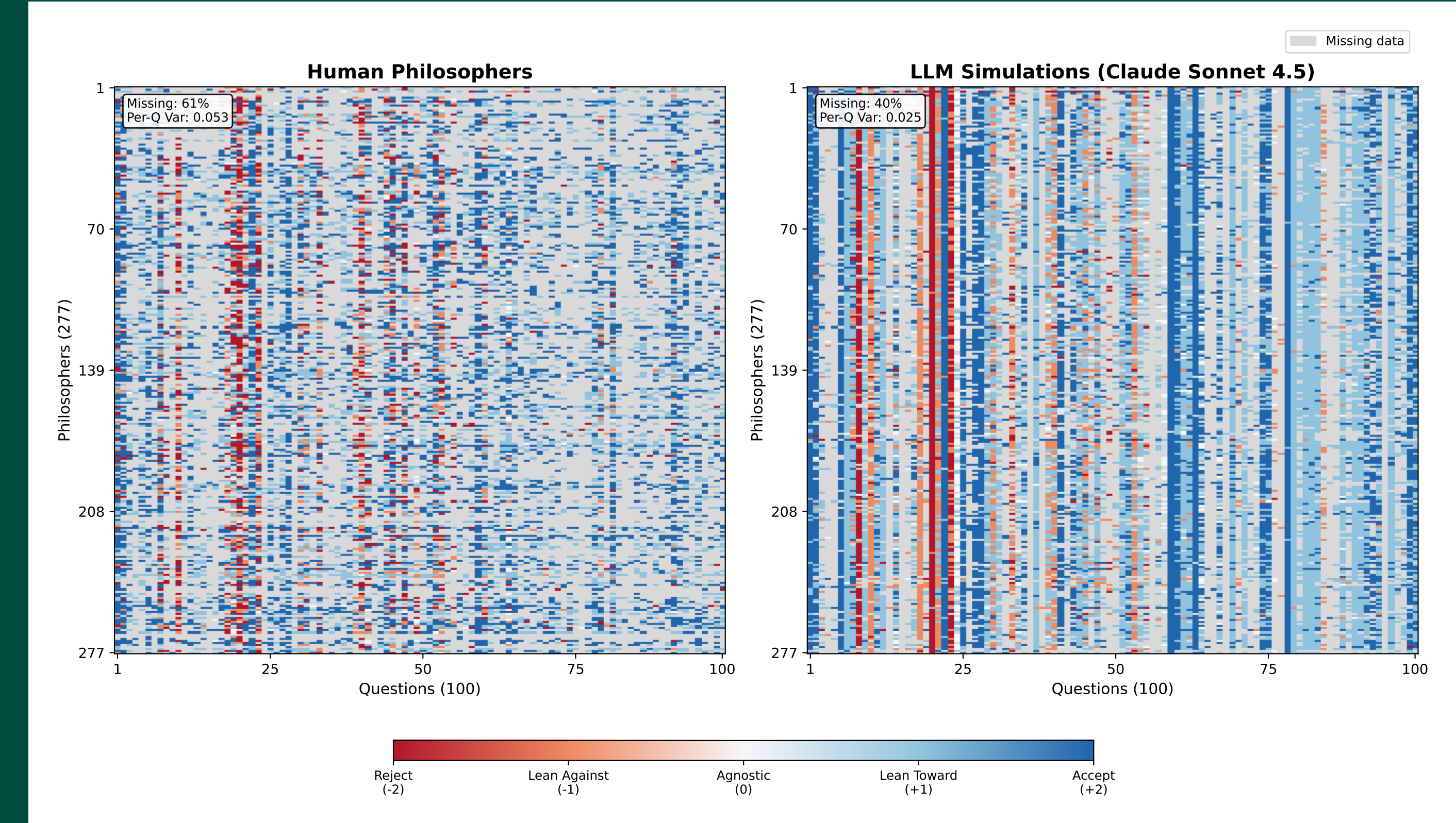
Source	Resp%	Per-Q Var
Human	38.9%	0.053
Claude-Sonnet-4.5	60.0%	0.026
Llama-3.1-8B	69.1%	0.026
Mistral-7B	67.7%	0.028
Llama-3.1-8B (FT)	66.5%	0.019
Qwen-3-4B	83.3%	0.016
GPT-4o	60.2%	0.014
GPT-5.1	59.9%	0.014

Humans disagree more than every silicon sample. Models fill in more answers, with less spread.

Normative domains fail most

- Models look accurate where philosophers already agree (aesthetics, language).
- They fail on applied ethics and philosophy of religion — the domains with the most human disagreement, and the ones that matter for alignment.
- Prediction error tracks human variance ($r = 0.75$): the model is not “better at ethics”; it just benefits from consensus questions.

A social world model is adequate only if it matches the distribution of human minds.



Each cell is one philosopher on one question. Humans are speckled; Claude forms vertical bands.

Silicon philosophers fail this test: they are too uniform, and they disagree along the wrong axes.



Take a picture for the paper and data explorer.

PCA: the axes of disagreement

Principal component analysis finds the main directions along which philosophers differ — following Bourget & Chalmers on the PhilPapers survey.

- **Var(6)** is the share of disagreement captured by the first 6 components. Higher than humans means the model collapsed onto fewer axes.
- **Elem r** asks: do the same question-pairs hang together as in humans?
- **Load r** asks: do the same questions load on the same components?

Human disagreement is multi-dimensional: five axes each explain $\geq 2\%$, and PC1 is naturalism / meta-ethics — not a single left–right spectrum. LLMs do not recover those axes. GPT models dump more variance into six components. Their PC1 clusters by surface topic (mind uploading, spacetime), not by the human cross-domain pattern. Even the best loading alignment stays weak.

Source	Var(6)	Elem r	Load r
Human	71.1%	—	—
Claude-Sonnet-4.5	72.3%	0.245**	0.023
GPT-4o	83.0%	0.103**	0.110
GPT-5.1	81.6%	0.099**	−0.137
Mistral-7B	73.8%	0.027	−0.041
Qwen-3-4B	67.8%	0.027	0.045
Llama-3.1-8B (FT)	61.2%	0.065*	−0.063
Llama-3.1-8B	62.4%	0.016	0.026

** $p < 0.01$, * $p < 0.05$.

Invented specialist links

Each silicon philosopher is prompted with a PhilPeople profile, including areas of specialization and areas of interest. Models treat those labels as a cue for the *question’s topic*, and invent a stance that “fits” the words.

- Philosophy of Biology in the prompt $\rightarrow$ the “biological” view of personal identity on the test item, because the words overlap. Real biology specialists do not show this link.
- Ancient philosophy in the prompt $\rightarrow$ “must be Aristotelian” on practical reason. Humans: no such link.
- The same lexical shortcut runs the other way: biology specialists are pushed *against* psychological identity.
- These are bogus relationships: large and significant in the silicon sample, absent in the 277-philosopher ground truth and in PhilPapers 2020.

Specialist effect	Human	LLM
	Diff Sig	Diff Sig
Biology $\rightarrow$ biol. identity	+11.4 pp n.s.	+43 pp 4/7
Biology $\rightarrow$ psych. identity	+4.1 pp n.s.	−65.7 pp 3/7
Ancient $\rightarrow$ Aristotelian	+1.9 pp n.s.	+68.9 pp 7/7

DPO fine-tuning improves question-to-question structure but does not restore human-like diversity.